Secure Engine Firmware Release Notes
For EFR32xG24 Products

Secure Engine (SE) firmware is installed with the product. The firmware is upgradable through Gecko Bootloader and Simplicity Studio. Features and security updates available in the latest release are described in this document.

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Using This Release

Secure Engine (SE) firmware is factory installed with the product.

SE firmware upgrade images are installed with Gecko SDK Suite in Simplicity Studio. This release contains the following.

- Upgrade image for Gecko Bootloader (.seu file)
- Upgrade application (.hex file)

Upgrade images are signed and encrypted.

Use of the upgrade image with Gecko Bootloader is described in *UG266: Silicon Labs Gecko Bootloader User's Guide, Section on Secure Engine Upgrade.*

To upgrade your device using the Upgrade application, program the .hex file to Flash using Simplicity Commander and then reset the device or use the upgrade feature in Simplicity Studio.

The Secure Engine Manager component in the Gecko SDK Suite provides an API to get the current SE version from the device. The API reference guide is available on https://docs.silabs.com/gecko-platform/latest/service/api/group-sl-se-manager

1.1 Compatible Products

The firmware is compatible with the following Series-2 products:

- EFR32xG24 SoCs

1.2 Support

Development Kit customers are eligible for training and technical support. You can use the Silicon Laboratories web site https://www.silabs.com/products/development-tools/software to obtain information about software products and services, and to sign up for product support.

You can contact Silicon Laboratories support at www.silabs.com/support

Features and Security Updates

v2.2.0

- Added support for TrustZone Root Key which can be used by TrustZone secure applications for secure storage. The key is renewed on OTP configuration, debug lock and device erase.
- Fixed a bug where TrustZone specific debug locks where not retained across soft resets after being opened through Secure Debug unlock.
- Fixed an issue where executing a host upgrade with a large image could cause a watchdog timeout.
- Security and stability updates

v2.1.7

- Fixed downgrade attack vulnerability
- Increased internal watchdog timeout to prevent long operations from being interrupted
- Make the Device Erase command break the boot loop that could occur after a failed host upgrade
- X25519 and Ed25519 algorithms, along with related key management functionality, is now supported on Secure Vault Mid devices
- Randomize the operating frequency between every command also when active mode is enabled
- GCM support for input lengths larger than 0.5 GB

v2.1.5

- Startup time improvements. For typical device configurations, this change could save 5-6 ms of boot time. With Secure Boot enabled and debug ports locked, the estimated boot time savings can be as much as 10 ms. Prior to this change, startup time could increase over time based on the history of the device. For example, performing a successful host upgrade or device erase command on a debug locked part could permanently increase boot time by a few milliseconds. In a worst-case scenario, these effects could add up to a total of 300 ms additional boot time.

v2.1.4

- Fixed a bug where using volatile keys together with a custom ECC curve did not work
- Added explicit validation of input and output length for commands that are sent without any input/output buffers from the mailbox interface. This causes a change in behavior of the SE Manager function `sl_se_roll_challenge`, which has been fixed in Gecko SDK v3.2.2. When applying the SE firmware upgrade to a device with an application compiled with an older version of the SDK, be aware that `sl_se_roll_challenge` will return an error code after the SE firmware upgrade, until the application is recompiled with the updated SDK.
- Improved fault injection hardening

v2.1.3

- Fixed bug where using custom domains failed for some commands

v2.1.2

- Added support for new glitch detectors through the tamper response bits in SM InitOTP
- Added PBKDF2-CMAC support (option 0x10) to DeriveKey command
- Added custom curve support in ECDSA and ECDH
- Improved fault injection resistance
- Security and stability updates

v2.1.1

- Production support update (Silicon Labs internal)

v2.1.0 (first release for EFR32xG24 products)
- Secure Vault High features

Known Issues

- None

Legal

4.1 Disclaimer

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4.3.1 t_cose

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4.3.2 QCBOR

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